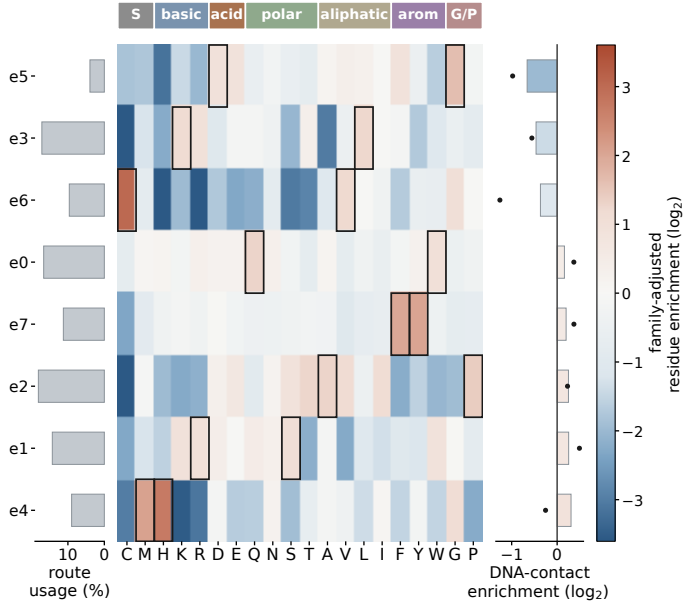
Mixture-of-experts separates structural scaffolding from DNA-recognition chemistry

experts ordered along the observed DNA-contact axis in every panel: scaffold-enriched → mixed → DNA-interface-enriched

a

Experts show distinct residue-chemistry signatures

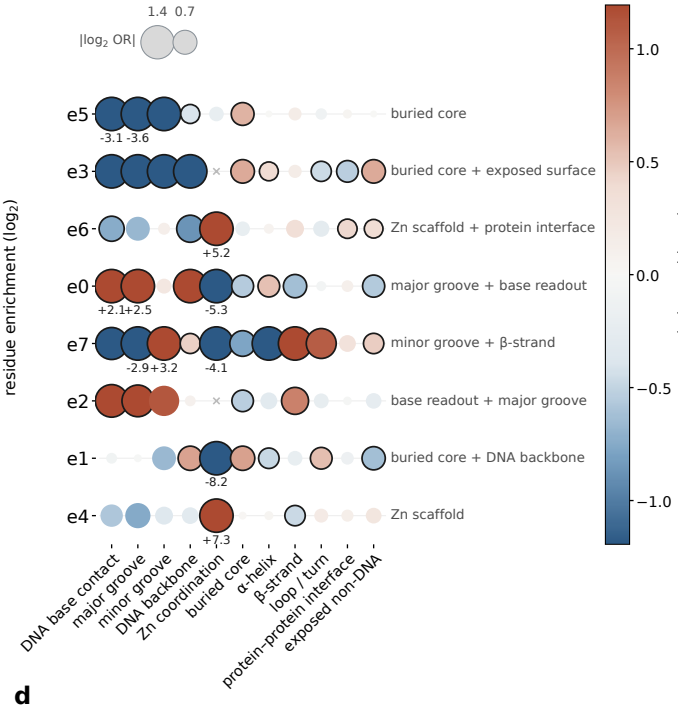
TF-family composition controlled; box = 2 strongest residues
bars = v23 supervision set, dots = recomputed co-crystal contacts



b

Experts separate structural scaffolding from DNA-interface functions

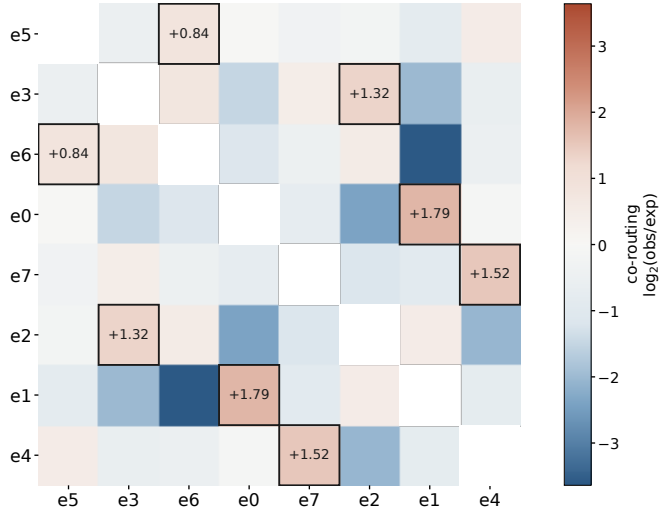
amino-acid identity and TF family both controlled; outline = FDR < 0.05
colour and area saturate at |log2 OR| = 1.2; saturated cells are labelled



c

Top-2 routing combines experts non-randomly

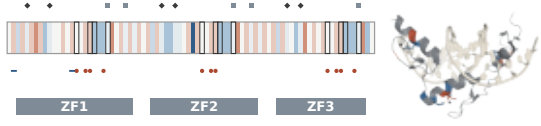
expectation preserves each expert's marginal usage
boxed = strongest enriched pairs



d

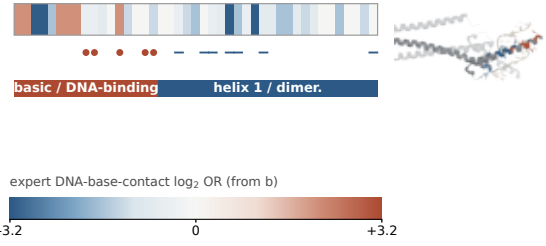
C2H2: metal-scaffold and base-readout programs

Egr1 · PDB 1P47



bHLH: helical-scaffold and basic DNA-contact programs

MYC · PDB 1NKP



◆ Zn–Cys ■ Zn–His ● DNA base contact ▬ protein–protein interface □ canonical recognition position